ENGINEERING & ARCHITECTURE DEEP DIVE

Comparative Analysis: Dropdown & Navigation Hierarchy Systems

Comprehensive evaluation of collapsible trees, tri-state ticking, completion cascade, viewport handling, and persistence in **Solid State Physics** vs. **LaTeX Interactive**.

Target Systems: SOLID_STATE_1234.html vs. LATEX_1.html **Source Engines:** tools/ss12/engine.js vs. tools/latex1/src/ui.js **Author/Recipient:** ihjas.one@gmail.com **Date:** October 2026

1. Executive Summary & Philosophy Matrix

Both applications implement a three-tiered hierarchical structure (**Module ▸ Section / Unit ▸ Topic**) with tri-state completion ticks and Firebase synchronization. However, their interaction models and UX philosophies diverge completely based on subject matter requirements:

SOLID STATE PHYSICS (TOOLS/SS12)

Multi-Open Exploratory Workbench

Designed for deep analytical physics. Users frequently need to cross-reference multiple derivations, band gap figures, and 3D crystal structures simultaneously. Ticking a section **preserves viewport position** and keeps the card open. Open/closed states are persisted across browser reloads.

LATEX INTERACTIVE (TOOLS/LATEX1)

Focus Accordion & Mastery Sprint

Designed for sequential tutorial learning and code editing. Enforces a **strict single-open focus accordion** per level. Marking a topic or section as completed **automatically collapses the card** and smoothly centers the header, giving students immediate psychological closure and a clear desk.

2. Comprehensive Technical Feature Matrix

System Dimension	Solid State Physics (SS12)	LaTeX Interactive (LX1)
Accordion Strategy	**Multi-card independent toggle**. Opening card *A* leaves cards *B* and *C* untouched. Multiple modules and units can be open simultaneously.	**Strict single-open focus accordion**. Opening card *A* immediately queries sibling nodes at that level and closes them (`$$('.'+lv+'.open').forEach(c=>setOpen(c,false))`).
Tick Click Action	**Toggle state only**. Ticking ✓ toggles `state.done[id]`. The dropdown stays completely open. Title color dims to `var(--dim)`.	**Toggle state + Auto-collapse + Scroll**. If marked done, the dropdown **instantly closes**, saves state, and triggers smooth auto-scroll to center the header.

System Dimension	Solid State Physics (SS12)	LaTeX Interactive (LX1)
Dedicated Footer Done Button	Modules III & IV contain a `.donebar` with a checkbox, but clicking it keeps the station in place without closing.	**Explicit ✓ `Mark as completed` CTA**. Located at the bottom of every topic. Closes the dropdown, scrolls to header, and displays a toast (`Completed ✓`).
Initial Load State	**Modules open, sections closed**. On first visit, top-level modules default to open. Subsequent visits restore whatever the user had opened.	**Everything collapsed by default**. Starts in a pristine zero-clutter state so the student actively chooses where to begin.
Open State Persistence	**Persisted in LocalStorage**. `state.open = { [cardId]: 1 }` is stored in `ss12_progress_v1`. Exactly restores open cards on page reload.	**Intentionally Ephemeral**. `state.open` is *never* persisted. Reloading the page returns to clean collapsed state.
Expand / Collapse All	**Global button available**. Header icon button (⇕ / ⇡) expands or collapses all modules, sections, and topics simultaneously.	**Omitted by Design**. An "Expand All" command conflicts with the single-card focus accordion model.
Viewport & Scrolling	**Manual scrolling**. Opening a card does not shift viewport scroll position.	**Intelligent smooth auto-scroll**. Calculates `getBoundingClientRect().top`. If off-screen (< 60px or > 45vh), smoothly centers the target.
Container Width Melting	**Fixed reading column**. Uses standard responsive column widths (`--col: 52rem`).	**Depth-aware width melting**. When an inner card is open, parents shed padding via `.has-open`. Wide code blocks trigger `document.body.classList.add('wide')`.
Canvas / Widget Lifecycle	**Canvas re-measuring hook**. Since collapsed cards have 0-width canvases, opening triggers `resizeFigsIn(card)` after 30ms to redraw 2D/3D plots.	**Lazy widget hydration**. Code snippets, Shiki syntax highlighters, and live editors (`[data-ex]`, `[data-w]`) hydrate only on open.
Bookmarks Integration	None. All navigation is bound directly to the course tree.	**Full Bookmarks mirror system**. Every dropdown has a ☆/★ toggle. Starred items dynamically populate a dedicated **Bookmarks Tab**.

3. In-Depth Architectural Comparison

3.1 Ticking Mechanism & Completion Cascades

Both implementations calculate progress via hierarchical leaf nodes. However, their reaction to completion is radically different:

Solid State (tools/ss12/engine.js):

```
function toggleNode(id) {
  const mod = TREE.find(m => m.id === id);
  if (mod) {
    const L = leavesOfMod(mod);
    setLeaves(L, !stateOf(L).all);
  } else {
    const sec = SEC[id];
    if (sec) {
      const L = leavesOfSec(sec);
      setLeaves(L, !stateOf(L).all);
    } else {
      setLeaves([id], !state.done[id]);
    }
  }
  persist();
  paintAll();
  // Card remains open! View does not shift.
}
```

LaTeX (tools/latex1/src/ui.js):

```
function toggleNode(id, src) {
  const L = leavesOf(id);
  const nowDone = !stateOf(L).all;
  setLeaves(L, nowDone);
  persist();

  // If marked done: close card & center header
  if (nowDone) {
    const c = cardOf(src, id);
    if (c) {
      setOpen(c, false);
      saveLocal();
      const h = c.firstElementChild || c;
      setTimeout(() => h.scrollIntoView({
        behavior: 'smooth', block: 'center'
      }), 50);
    }
  }
}
```

> **Key Observation on User Experience:** In **LaTeX**, when a student finishes working through Tutorial III and taps *"✓ Mark as completed"*, the tutorial smoothly folds up, removing the cognitive load of that completed section, and highlights the next task. In **Solid State**, students frequently refer back to equations, Bragg planes, and Fermi velocity values while tackling subsequent derivations, so keeping cards open avoids disruptive re-clicks.

3.2 Focus Accordion vs. Multi-Open Tree

LaTeX enforces strict single-element activation using the `openCard` coordinator:

```
function openCard(card, opt) {
  opt = opt || {};
  const lv = levelOf(card);
  const root = card.closest('.view') || document;

  // 1. Close all sibling cards at the same hierarchy level
  $$('.' + lv + '.open', root).forEach(c => {
    if (c !== card) setOpen(c, false);
  });

  // 2. Ensure all ancestor levels (module, section) are open
  let p = card.parentElement && card.parentElement.closest('.mod,.unit');
  while (p) {
    if (!p.classList.contains('open')) {
      $$('.' + levelOf(p) + '.open', root).forEach(c => {
        if (c !== p) setOpen(c, false);
      });
      setOpen(p, true);
    }
    p = p.parentElement && p.parentElement.closest('.mod,.unit');
  }

  // 3. Open the target card
  setOpen(card, true);
  saveLocal();

  // 4. Smooth scroll header into clear view
  if (opt.scroll !== false) {
    const h = card.firstElementChild;
    setTimeout(() => {
      const top = h.getBoundingClientRect().top;
      if (top < 60 || top > innerHeight * 0.45) {
        h.scrollIntoView({ behavior: 'smooth', block: 'start' });
      }
    }, 340);
  }
}
```

3.3 Depth-Aware Layout & Width Melting

A standout architectural feature in LaTeX's dropdown engine that does not exist in Solid State is **Width Melting**. In deeply nested documentation trees, nested borders and paddings normally eat away mobile horizontal space. LaTeX solves this dynamically:

- **`.has-open` Class:** When a child section or topic opens, its parent `.mod` or `.unit` sheds its internal padding so the child can expand edge-to-edge.
- **`body.wide` Toggle:** When an opened dropdown contains wide interactive code widgets (`.wg`) or editable Shiki playgrounds (`.ex`), `markDepth()` adds `.wide` to `<body>`, instantly widening the main layout container from 54rem to 76rem.

3.4 Visual Style & Component Anatomy

Component Element	Solid State Physics Implementation	LaTeX Interactive Implementation
Tick Box (.tick)	Border `1.8px solid var(--rule)`, radius `0.36rem`. Partial state: 45° diagonal slice via `linear-gradient(135deg, ...)`. Full state: Solid fill with theme copper/teal/crimson accent.	Border `1.5px solid var(--rule)`, radius `0.4rem`. Partial state: CSS pseudo-element `::after { content: '–'; }`. Full state: CSS pseudo-element `::after { content: '✓'; }` in accent color.
Caret / Chevron	Character ► rotated via CSS `transform: rotate(90deg)` when parent has `.open` class.	Character ► in headers, plus dedicated smooth SVG rotating chevron (`.chev`) inside `details.qa` question accordions.
Exam Q&A Dropdowns	Custom `.check` retrieval boxes with 1-5 confidence rating chips + reveal button; native `<details class="prob">`.	Standardized `<details class="qa">` with question badge (`.pn`), title (`.pt`), animated chevron, and bookmark star.
Completed Header Style	Header text color shifts to muted gray (`.unit.done > .uh h3 { color: var(--dim); }`).	Header text dims to `var(--dim)`, plus footer CTA transitions to a soft green pill (`.btn.completed` with green border & background).

4. Synthesis & Recommendations

> **Can these systems be cross-pollinated?**
> Yes. The two designs have evolved to suit their specific subject matter, but several high-value features can be transferred between them.

1. **What Solid State could adopt from LaTeX:**
   - **Smooth Auto-Scroll on Open:** Long topics in Module I and II (e.g. *Brillouin Zones*) often open with the header scrolled half out of view. Adding LaTeX's `scrollIntoView({ behavior: 'smooth', block: 'start' })` would improve navigation.
   - **Footer "Mark as Done" CTA:** Adding an explicit completion button at the end of long derivations helps students signal completion without having to scroll all the way back to the top of the card.
   - **Bookmarks Tab:** Solid State physics has 37 extensive derivations; allowing students to star difficult ones (e.g. *Kronig-Penney*, *Curie-Weiss Law*) into a quick-access tab would be valuable for revision.
2. **What LaTeX could adopt from Solid State:**
   - **Optional "Expand All" / Review Mode:** For rapid pre-exam cramming, a toggle to view all cheat-sheet notes simultaneously without auto-closing would be useful.
   - **Canvas / Dynamic Element Resize Handling:** Solid State's `resizeFigsIn()` delayed re-measure pattern is the gold standard for preventing zero-width SVG or canvas rendering bugs in collapsible layouts.